CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating and Evidence-Grounded Reasoning for Trustworthy Fake News Triage

LEGAL & METHODOLOGICAL DISCLAIMER: *A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify whether a news story is factually true. URL extraction, model pattern recognition, and external fact verification remain strictly separate processes. High uncertainty cases are flagged for professional human editorial review.*

Source / URL: Manual Text Submission
Title: Senate Approves Major Clean Energy Deal
Word Count: 28 words | **Date:** 2026-10-19 11:42:49

VERDICT: UNCERTAIN

Calibrated Confidence: 87.0% (Threshold tau = 80.0%)
Base Disagreement (std): 0.236 (Threshold delta = 0.20)
Gating Trigger: High Disagreement

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Model Vote	Confidence
Random Forest	57.6%	FAKE	57.6%
XGBoost	38.6%	REAL	61.4%
LightGBM	58.3%	FAKE	58.3%
Logistic Regression	0.1%	REAL	99.9%
CAGE-Fake Stack	13.0%	REAL	87.0%

Consensus Status: High Dissent / Conflicting Predictions
Base Disagreement (4-base std): 0.236 (Delta = 0.20)
5-Model Consensus Spread: 0.235
Vote Distribution: 2 FAKE vs 3 REAL
Diagnostic: Base-model disagreement (0.236) exceeds safety threshold delta (0.200).

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	77.4/100	Adequate
Source Attribution	50.9/100	Deceptive / Alert
Epistemic Grounding	77.4/100	Adequate
Punctuation Discipline	80.3/100	Excellent
Readability Structure	77.4/100	Adequate
Lexical Diversity	62.1/100	Adequate

Overall Credibility Index: 69.5/100
Credibility Tier: B (Standard News Distribution)

Scores are normalized against verified newsroom style guides: high attribution, balanced syntax, and low sensationalism maximize the index.

3. Transparent Reasoning & Counterfactual Sensitivity

Diagnostic Explanation:
1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **87.0% calibrated confidence**.
- **Gating Trigger (High Disagreement)**: Base-model disagreement (0.236) exceeded threshold delta (0.200). Although confidence was 87.0%, the base classifiers sharply diverged in their verdicts. The system abstains to ensure decisions require multi-model consensus.

Counterfactual Simulation: No qualifying sensational words, speculative hedges, or shout punctuation detected in the article text; no counterfactual perturbations were applied.
Original P(FAKE): 13.0% | **Transformed P(FAKE):** 13.0% **(Signed Shift: +0.0 percentage points)**

4. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Weight / Impact	Top REAL-Driving Anchors	Weight / Impact

Random Forest Prediction [P(FAKE)]	+0.191	Logistic Regression Prediction [P(FAKE)]	-2.193
LightGBM Prediction [P(FAKE)]	+0.080	XGBoost Prediction [P(FAKE)]	-0.191
-	-	Inter-Model Base Disagreement (std)	-0.072

5. Empirical Benchmark Model Comparison (Held-out Test Split)

Model	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest	0.8639	0.8635	0.1548	100.0%	0.8639
XGBoost	0.8770	0.8769	0.0380	100.0%	0.8770
LightGBM	0.8689	0.8688	0.0418	100.0%	0.8689
Logistic Regression	0.8967	0.8967	0.0381	100.0%	0.8967
Uncalibrated Stack	0.8984	0.8983	0.0330	100.0%	0.8984
CAGE-Fake (Calibrated + Gated)	**0.9000**	**0.8999**	**0.0362**	**84.4%**	**0.9476**

6. Research Calibration & Disagreement Visualizations







